RELATIONSHIPS BETWEEN BOND PRICES AND BOND FEATURES

Learning Outcome

- identify the relationships among a bond's price, coupon rate, maturity, and yield-to-maturity

Inverse Relationship

As seen in the prior lesson, the price of a fixed-rate bond will change as the interest rate used to discount future cash flows changes. For any time-value-of-money calculation, a higher discount rate results in a lower present value for any fixed future cash flow and a lower discount rate results in a higher present value. Therefore, bond yields-to-maturity and prices move in opposite directions.

Coupon Effect

The *size* of bond coupon cash flows affects how much a bond's price will change for a given yield change for bonds of the same maturity. The lower a bond's coupon, the higher the *proportion* of total cash flow that occurs at maturity. Discounting this final cash flow by the factor of $(1 + r)^N$ as shown in Equation 2 magnifies the impact of a given change in r relative to bonds whose final cash flow is a smaller proportion of the total. This coupon effect is best demonstrated by comparing the extreme case of a zero-coupon bond (with 100% of cash flows including cumulative interest paid at maturity) with an otherwise identical fixed-coupon bond, as in the following example.

EXAMPLE 5

Yield Change Effect for Zero-Coupon vs. Coupon Bond

Assume an investor has the choice between the 4.625% fixed-coupon 30-year Romania Eurobond priced at par and an otherwise identical zero-coupon bond from the same issuer at the same yield-to-maturity of 4.625%.

Calculate the percentage change in price for both bond alternatives if YTM rises or falls by 100 bps upon issuance.

Use Equation 1 (zero-coupon bond) and the Excel or Google Sheets PV function (4.625% coupon bond) to calculate and compare these changes. Please refer to the candidate learning ecosystem online for a spreadsheet of these calculations in the downloaded Microsoft Excel workbook.

Face Value	100
Periods / year	1
Number of periods	30
Coupon payment	4.625

Change	YTM	Prices		Price Changes		Percentage Price Changes	
		Coupon	Zero	Coupon	Zero	Coupon	Zero
--	0.04625	100.000	25.759	--	--	--	--
0.01	0.05625	85.665	19.364	–14.335	–6.395	–14.34%	–24.83%

Rate Your Confidence
High
Medium
Low
Continue »
Category
Fixed-Income Bond Valuation: Prices and Yields
Related Questions:
Practice questions related to this topic

For the same time-to-maturity, a lower-coupon (or zero-coupon) bond has a greater percentage price change than a higher-coupon bond when market discount rates change by the same amount.

Maturity Effect

The time-to-maturity of a bond also affects a bond's price/yield relationship. Generally, all else equal, a longer-term bond has a greater percentage price change than a shorter-term bond when the market discount rates change by the same amount. The maturity effect is the result of the higher *N* in Equation 2 for the longer-maturity bond.

EXAMPLE 6

Yield Change Effect for Different Maturities

Assume an investor has a choice between Bond A, with a 1.5% fixed coupon and a 30-year maturity priced at par, and Bond B, a 10-year, 1.5% fixed-coupon bond priced at par from the same issuer.

Calculate the percentage change in price for these if YTM rises or falls by 100 bps upon issuance.

Compare the price changes in the 10-year and 30-year bonds using the PV function to calculate and compare these changes. Please refer to the candidate learning ecosystem online for a spreadsheet of these calculations in the downloaded Microsoft Excel workbook.

Bond	A	B
Face Value	100	100
Price	100	100
Coupon Rate	0.015	0.015
Coupon Payment	1.50	1.50
Maturity	30	10
100 bps rise in YTM		
YTM	0.025	
Price	79.070	91.248
% change	–20.93%	–8.75%
100 bps fall in YTM		
YTM	0.005	
Price	127.794	109.730
% change	27.79%	9.73%

We see that the percentage price change for the longer-maturity bond is larger than that for the shorter-maturity bond. Also, we see that, in price terms, the change for a YTM *decrease* is larger than that for a YTM *increase*.

Exceptions to the maturity effect exist but are rare in practice. They occur only for low-coupon (but not zero-coupon), long-term bonds trading at a discount. The maturity effect always holds on zero-coupon bonds, as it does for bonds priced at par value or at a premium above par value.

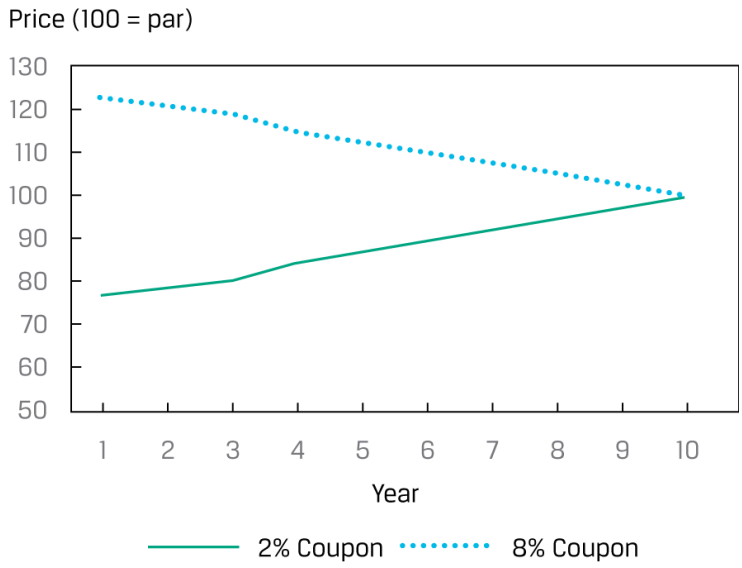
Constant-Yield Price Trajectory

Bond prices change with the passage of time, even if the market discount rate remains the same. As time passes, so as *N* in Equation 2 approaches zero, the bondholder comes closer to receiving the par value at maturity, provided the issuer does not default. The

premium. At each time, the bond price is recalculated using a market discount rate of 5%.

Exhibit 6:

Constant-Yield Price Trajectory



▶ Expand / collapse Extended Description

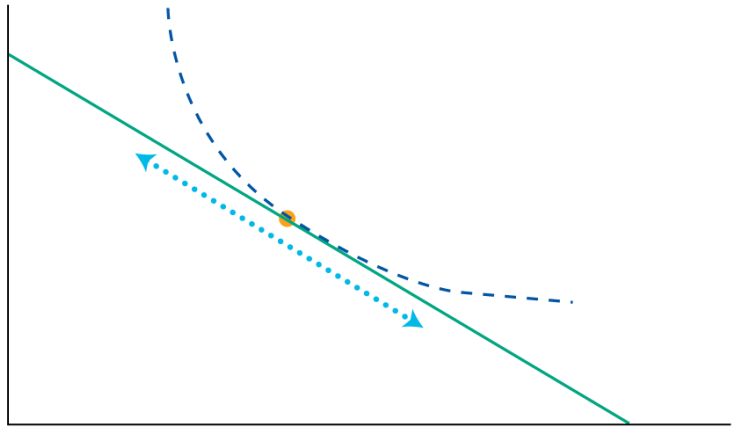
The 2% coupon bond's initial price is 76.835 per 100 of par value. The price increases each year and approaches par value as the maturity date nears. The 8% coupon bond's initial price is 123.165, and it decreases each year, approaching par value as the maturity date nears. Both prices are "pulled to par." Please refer to the candidate learning ecosystem online for a spreadsheet of these calculations in the downloaded Microsoft Excel workbook.

Convexity Effect

The *percentage* change in a bond's price will also vary depending on how the yield changes. The percentage price *increase* is greater, in absolute value, than the percentage price *decrease*. This implies that the relationship between bond prices and yields is not linear; instead, it is curved and "convex." Exhibit 7 shows this general relationship between a bond's price and its yield-to-maturity.

Exhibit 7:

Convex Price–Yield Relationship



Price

EXAMPLE 7

BRWA Bond: Convex Relationship

Despite a uniform increase and decrease in the BRWA bond's annual yield-to-maturity, notice how the resulting changes in price are different. Please refer to the candidate learning ecosystem online for a spreadsheet of these calculations in the downloaded Microsoft Excel workbook.

Face Value	100
Coupon payment	1.6
Number of periods	10

YTM Change	YTM	Price	% Price Change
--	0.016	100.00	--
0.004	0.020	96.41	–3.59%
–0.004	0.012	103.75	3.75%

A decrease in the yield-to-maturity from 160 bps to 120 bps results in a bond price increase of 3.75%, but an increase in the YTM from 160 bps to 200 bps results in a *smaller* bond price decrease. Based on this, we can say that the BRWA bond has *positive* convexity. Convexity will be discussed in greater detail in a later lesson.

Knowledge Check

1. Recall the Romania bond from Example 4. It pays annual coupons at a rate of 4.625% and has a face value of 100. Its issue date is 3 April 2019 and its maturity date is 3 April 2049. Calculate its original yield-to-maturity at issue if its issue price is 99.488.

Hide Solution

Solution:

The original yield-to-maturity is 4.657%. The YIELD function arguments in Microsoft Excel or Google Sheets are as follows:

```
=YIELD(SETTLEMENT, MATURITY, RATE, PR, REDEMPTION, FREQUENCY,[BASIS])
```

=YIELD(DATE(2019,4,3),DATE(2049,4,3),0.04625,99.488,100,1)

2. Describe the coupon effect.

Hide Solution

Solution:

Given two non-callable, fixed-coupon bonds that have different coupons but are otherwise identical, for a change in yield, the lower-coupon bond will have a greater percentage price change than the higher-coupon bond.

3. Describe the maturity effect.

Hide Solution

Solution:

Generally, for two bonds that are identical except for their times-to-maturity, the longer-maturity bond will have a larger percentage price change than will the shorter-maturity bond when yields change by the same amount.

4. Two non-callable, fixed-coupon bonds have the same maturity date. One trades at a discount; the other trades at a premium. If the yield stays constant until the bonds mature and neither defaults, the two bonds' prices will, over time:

A. converge.
B. remain unchanged.
C. diverge.

Hide Solution

Solution:

A is correct. The prices will converge because of the "pull-to-par" effect. As the maturity date nears, the present values of future cash flows will fall or rise to the par value, provided that the issuer does not default.

Discuss Filter

Discussion

What do you want to discuss... (required)

TP why does the solution gives us an excel formula?? im not aware we are allow to use excel in the exam

Created 7 months ago by Tito Pizarro 2 replies | Last Activity: 5 months ago

Show All replies Reply to this Comment

JP we are not allowed to use excel formula

Created 5 months ago by Jean Paul AL Kharrat 0 replies | Last Activity: 5 months ago

Reply to this Comment

© 2025 CFA Institute. All Rights Reserved.

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search